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**MARMARA  
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**FACULTY OF ENGINEERING**  
**COMPUTER ENGINEERING DEPARTMENT**

**CSE4078 Introduction to Natural Language Processing Project**

Advanced LLM Evaluation Metrics Report

**Project Task**

Conversation

**Group#7 Members**

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# Introduction

In this report, we present the evaluation of our base and fine-tuned models using advanced evaluation metrics. We build upon the metrics used in Delivery#3 and explore additional methods to gain a deeper understanding of the model performance.

Mover Score is a robust evaluation metric that has a profound impact on the development of text generation systems.

The BERT score is a metric used to evaluate the similarity between sentences by comparing their contextual embeddings. It measures how well a predicted sentence matches a reference sentence, considering semantic meaning rather than just surface-level word overlap.

The BLEURT score is a metric for evaluating text generation by leveraging pretrained models to assess the quality of sentences. It provides a more nuanced comparison by incorporating contextual and semantic information, making it effective for tasks like machine translation and text summarization.

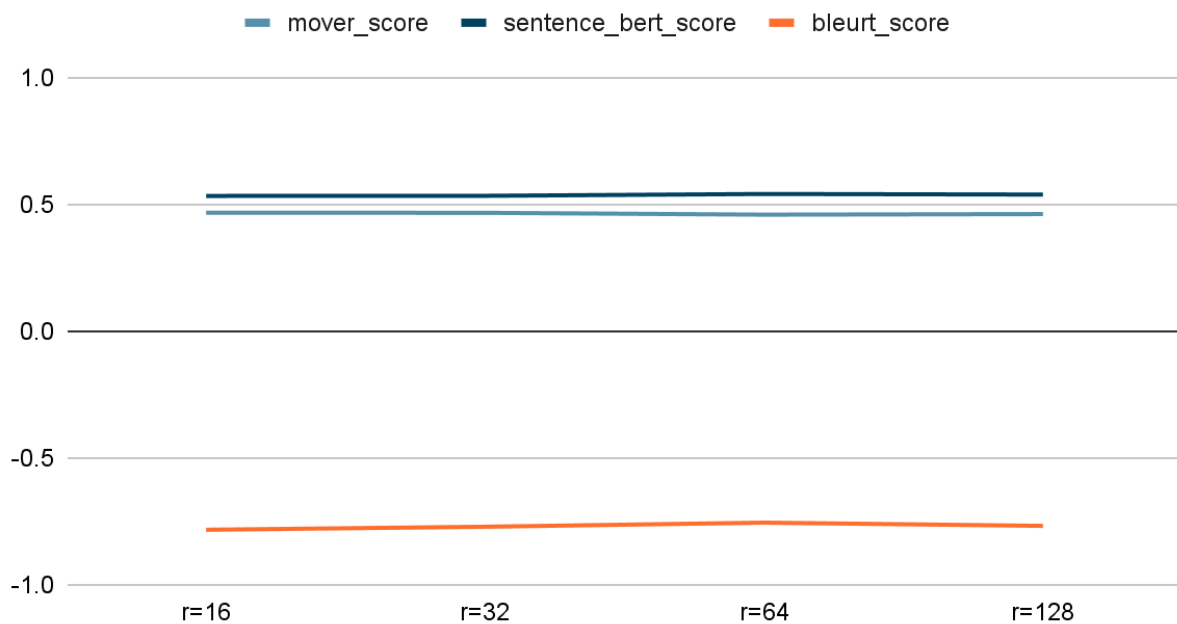
## Zero Shot Advanced Metrics

| rank    | mover_score    | sentence_bert_score | bleurt_scores  |
|---------|----------------|---------------------|----------------|
| r = 16  | Avg = 0,59964  | Avg = 0,40035       | Avg = -1,21417 |
|         | Std = 0,24211  | Std = 0,24211       | Std = 0,618947 |
| r = 32  | Avg = 0,59964  | Avg = 0,40035       | Avg = -1,21417 |
|         | Std = 0,24211  | Std = 0,24211       | Std = 0,61894  |
| r = 64  | Avg = 0,599648 | Avg = 0,400352      | Avg = -1,21417 |
|         | Std = 0,242113 | Std = 0,242113      | Std = 0,618947 |
| r = 128 | Avg = 0,599648 | Avg = 0,400352      | Avg = -1,21417 |
|         | Std = 0,242113 | Std = 0,242113      | Std = 0,618947 |

## One Shot Advanced Metrics

| rank           | mover_score   | sentence_bert_score | bleurt_scores  |
|----------------|---------------|---------------------|----------------|
| <b>r = 16</b>  | Avg = 0,46690 | Avg = 0,53309       | Avg = -0,78272 |
|                | Std = 0,15238 | Std = 0,15238       | Std = 0,37628  |
| <b>r = 32</b>  | Avg = 0,46646 | Avg = 0,53353       | Avg = -0,77082 |
|                | Std = 0,16045 | Std = 0,16045       | Std = 0,37473  |
| <b>r = 64</b>  | Avg = 0,45920 | Avg = 0,54079       | Avg = -0,75466 |
|                | Std = 0,15535 | Std = 0,15535       | Std = 0,36434  |
| <b>r = 128</b> | Avg = 0,46154 | Avg = 0,53845       | Avg = -0,7674  |
|                | Std = 0,15908 | Std = 0,15908       | Std = 0,37341  |

## Points scored



Changes on the value of r variable is not causing any change on mover score, sentence bert score and bleurt score.